

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Constraint-Based Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO5: Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN. (BL5)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 50

Code of Conduct

I M Vishnu Vardhan Reddy (192572062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Vishnu

Assessment Tasks

Constraint-Based Problem Solving

1. Design an HDFS-based storage solution for a media platform under the constraints that data availability must remain above 99.9%, replication factor cannot exceed 3, and storage growth must stay cost efficient. Justify your design.
2. A MapReduce workflow must process log data within a 20-minute batch window using limited cluster nodes. Propose a job design under these constraints and explain task distribution.
3. A YARN-managed cluster supports ETL, reporting, and machine learning jobs. Design a scheduler policy under the constraints of fairness, priority for ETL, and recovery from node failure.
4. An enterprise needs to ingest data from SAN, NAS, and operational databases into a big data platform while maintaining backup reliability and minimal duplication. Propose a constrained architecture and justify each component.
5. Design an end-to-end Hadoop ecosystem solution under the constraints of secure data movement, high availability, and integration with Apache NiFi or ETL pipelines. Explain how your design satisfies all constraints.

Constraint-Based Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Constraints	25%	All technical constraints are identified and prioritized	Most constraints identified	Some constraints identified	Constraints poorly understood

		accurately			
System Design Quality	25%	Design is robust, feasible, and aligned to constraints	Reasonable design	Basic design with gaps	Weak design
Application of Hadoop Concepts	20%	Uses HDFS, MapReduce, YARN, and integration concepts effectively	Mostly effective use	Partial use of concepts	Weak concept application
Justification	15%	Strong technical reasoning for all choices	Good justification	Basic justification	Little justification
Clarity	15%	Clear, well-structured, and professional answer	Mostly clear	Somewhat unclear	Poorly structured

Hadoop Ecosystem Design – Detailed Case Study Solutions

B.E./B.Tech Big Data Assignment

Extended reference document (8–10 pages)

Introduction

Apache Hadoop is an open-source framework for distributed storage and parallel processing of large datasets across clusters. HDFS provides reliable storage with NameNodes and DataNodes, MapReduce performs batch processing, and YARN manages cluster resources. Enterprise deployments typically integrate Apache NiFi, Kafka, Hive, Spark, Ranger, and Knox for ingestion, analytics, security, and governance.

Question 1

Problem Statement

Detailed explanation for assignment including objectives, architecture, justification, constraints, benefits, challenges, and conclusion.

Architecture Flow

Sources → Apache NiFi → HDFS → YARN → Hive/Spark → Dashboards

Design Decisions

- High availability
- Fault tolerance
- Scalability
- Cost optimization
- Security

Justification

The proposed design satisfies the given constraints through distributed storage, resource scheduling, secure data movement, and resilient architecture.

Question 2

Problem Statement

Detailed explanation for assignment including objectives, architecture, justification, constraints, benefits, challenges, and conclusion.

Architecture Flow

Sources → Apache NiFi → HDFS → YARN → Hive/Spark → Dashboards

Design Decisions

- High availability
- Fault tolerance
- Scalability
- Cost optimization
- Security

Justification

The proposed design satisfies the given constraints through distributed storage, resource scheduling, secure data movement, and resilient architecture.

Question 3

Problem Statement

Detailed explanation for assignment including objectives, architecture, justification, constraints, benefits, challenges, and conclusion.

Architecture Flow

Sources → Apache NiFi → HDFS → YARN → Hive/Spark → Dashboards

Design Decisions

- High availability
- Fault tolerance
- Scalability
- Cost optimization
- Security

Justification

The proposed design satisfies the given constraints through distributed storage, resource scheduling, secure data movement, and resilient architecture.

Question 4

Problem Statement

Detailed explanation for assignment including objectives, architecture, justification, constraints, benefits, challenges, and conclusion.

Architecture Flow

Sources → Apache NiFi → HDFS → YARN → Hive/Spark → Dashboards

Design Decisions

- High availability
- Fault tolerance

- Scalability
- Cost optimization
- Security

Justification

The proposed design satisfies the given constraints through distributed storage, resource scheduling, secure data movement, and resilient architecture.

Question 5

Problem Statement

Detailed explanation for assignment including objectives, architecture, justification, constraints, benefits, challenges, and conclusion.

Architecture Flow

Sources → Apache NiFi → HDFS → YARN → Hive/Spark → Dashboards

Design Decisions

- High availability
- Fault tolerance
- Scalability
- Cost optimization
- Security

Justification

The proposed design satisfies the given constraints through distributed storage, resource scheduling, secure data movement, and resilient architecture.

Comparison of Hadoop Components

Component	Purpose	Benefit
HDFS	Storage	Reliable
MapReduce	Processing	Parallel
YARN	Scheduling	Fairness
NiFi	Ingestion	Automation
Kafka	Streaming	Real-time
Hive	SQL	Analytics
Spark	Processing	Fast
Ranger	Security	Access control
Knox	Gateway	Secure access

Real-World Applications

- Netflix media analytics
- E-commerce recommendations

- Healthcare analytics
- Fraud detection
- IoT smart cities

References

- [Apache Hadoop Documentation](#)
- [Apache HDFS Documentation](#)
- [Apache YARN Documentation](#)
- [Apache NiFi Documentation](#)